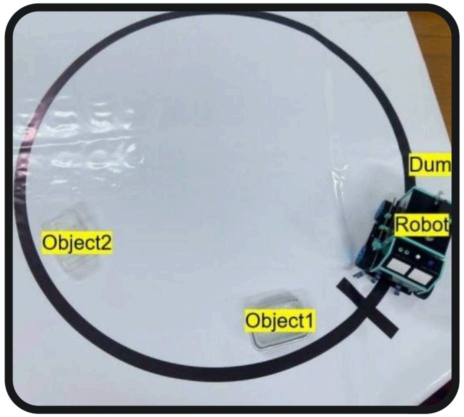
CONCLUSIONS

- In conclusion, the Automatic Dustbin System with Line Following Robot was successfully designed, implemented, and tested.
- The project showcased the practical application of systems and robotics in daily tasks such as waste disposal.

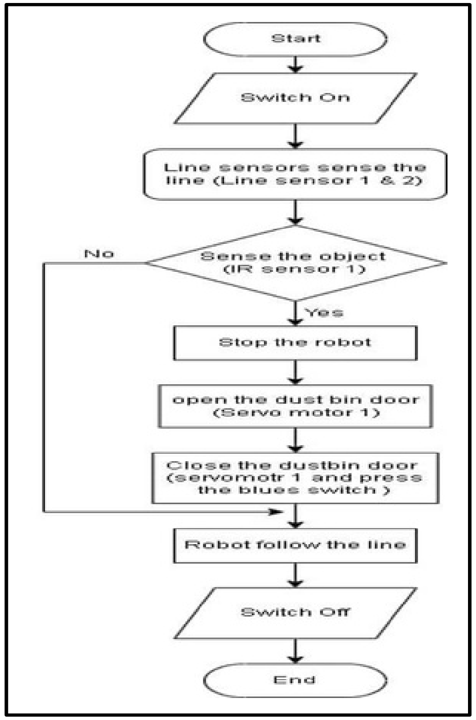


APPLIED AREAS

- Restaurants
- Healthcare and Hospital Environments
- Smart Homes
- Automated Cleaning Systems in Airports , Malls and Hotels.

SYSTEM PROCESS

- Step 1 : Start
- Step 2 : Switch On
- Step 3 : Line Tracking sensor 1 & 2 sense the line
- Step 4 : IR sense the object
- Step 5 : Stop the robot
- Step 6: Open the lid of the dustbin
- Step 7: Wait 3 seconds
- Step 8 : Close the lid
- Step 9: Robot follow the line
- Step 10:Switch Off
- Step 11:End



System Flowchart of the Line Following Mobile Robot in Specific Area

PROJECT SHOW & JOB FAIR 2026

ISO 9001:2015

TECHNOLOGICAL UNIVERSITY (MANDALAY)

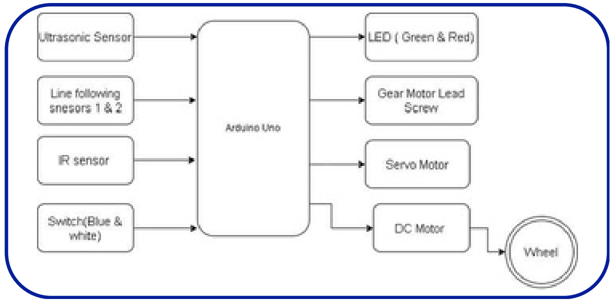
DEPARTMENT OF MECHATRONIC ENGINEERING

Smart Dustbin Mobile Robot with Line Following System

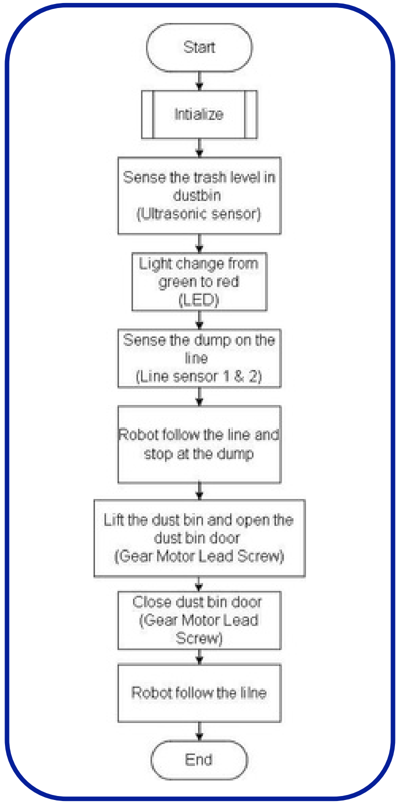
AIMS AND OBJECTIVES

To develop the automatic dustbin using line following robot inspecific areas.

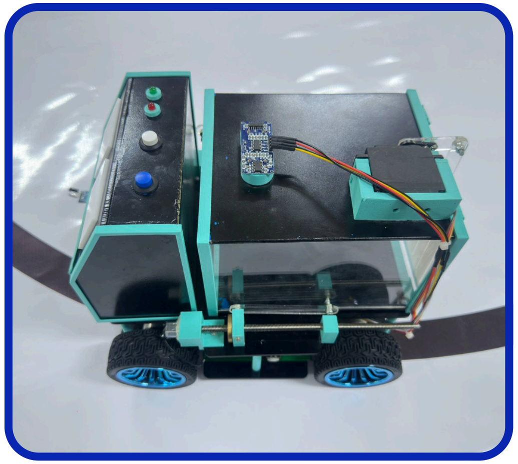
- Implement a line following robot using Arduino that can detect the objects and follow a predefined path.
- Collect waste automatically and dispose of it when the dustbin becomes
- full by lifting and dumping the dustbin container.
- Reduce the human intervention for waste-related tasks.



Overall Block Diagram Of The System



System Flowchart of Dustbin Level and Go to the Garbage Dump



HERDWARE COMPONENT

- Arduino Uno
- Two-Line Tracking Sensors
- IR Sensor
- HCSR04 Ultrasonic Sensor
- Four Wheels
- LM2596 DC to DC Step-Down Converter
- L298N Motor Driver Module (Dual H-Bridge)
- MG996R Servo Motor
- BMS Lithium Battery Protection Board
- Gear Motor with Lead Screw
- 8mm Shaft Rod
- LM8UU Linear Ball Bearing
- Three 18650 Batteries (Rechargeable)
- Two LEDs (Red and Green)
- Two Push Buttons (Blue and White)